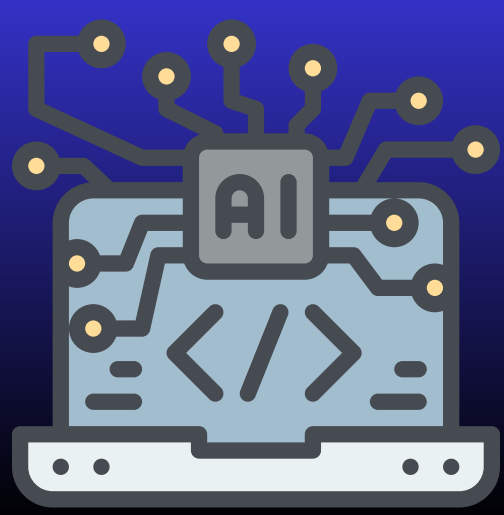




AI-Based Greenwashing Detection in Corporate Sustainability Reports Using NLP

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1. INTRODUCTION

Greenwashing refers to the practice of exaggerating or misrepresenting environmental performance through sustainability disclosures. As Environmental, Social, and Governance (ESG) reporting becomes increasingly important, organizations may use sustainability language that is not fully supported by measurable actions. This study proposes an AI-based framework that combines NLP, ESG risk indicators, topic modeling, and explainable machine learning to identify potential greenwashing behavior in corporate sustainability reports.

2. OBJECTIVES

- Analyse linguistic characteristics across ESG risk levels.
- Examine the relationship between environmental language density and ESG risk.
- Identify dominant sustainability disclosure themes using topic modelling.
- Develop a Greenwashing Risk Index (GRI) by integrating NLP features with ESG risk metrics.
- Build an explainable machine learning model to predict greenwashing risk.

3. DATA OVERVIEW

Two datasets were combined for the analysis:

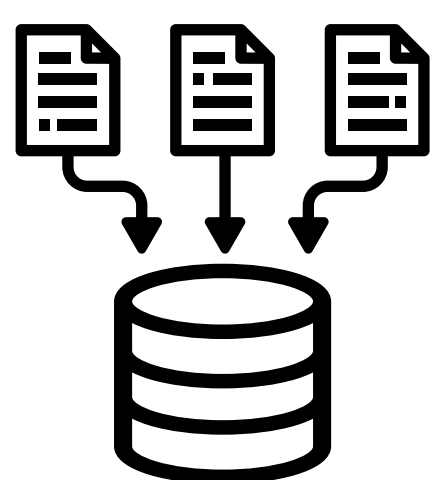
- Sustainability Reports Dataset: 866 corporate sustainability reports.
- ESG Risk Dataset: 503 companies with ESG risk indicators.
- Common Companies Matched: 262.
- Final Merged Dataset: 862 report-ESG pairs.

KEY FINDINGS

- High ESG-risk firms use more hedging language.
- Environmental disclosure intensity does not correlate with lower ESG risk.
- Sustainability and emissions management dominate corporate disclosures.
- The proposed Greenwashing Risk Index identifies high-risk firms.
- Explainable AI reveals ESG Risk Score and linguistic indicators as key predictors.

4. PRAPOSED METHEDODOLOGY WORKFLOW

1 Data Collection



Sustainability Reports
+
ESG Risk Dataset

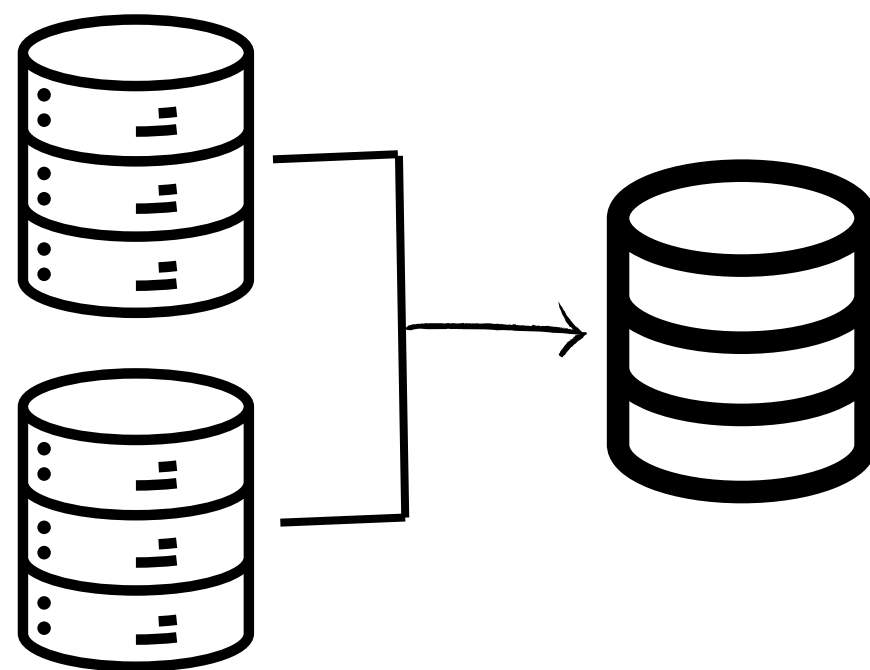
2 Text Preprocessing



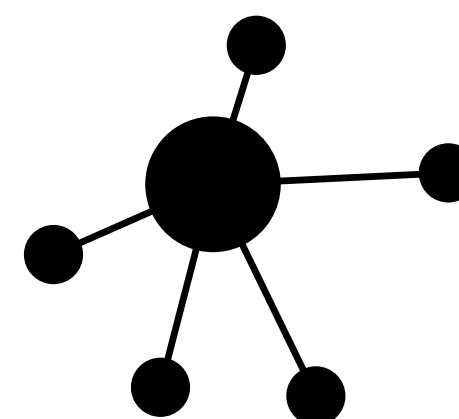
3 Feature Engineering (Linguistic Features)

- Hedging Language count
- Promotional Language Count
- Environmental Keyword Count
- Environmental language density
- Word Count

4 Data Integration



5 Topic Modeling BERTopic



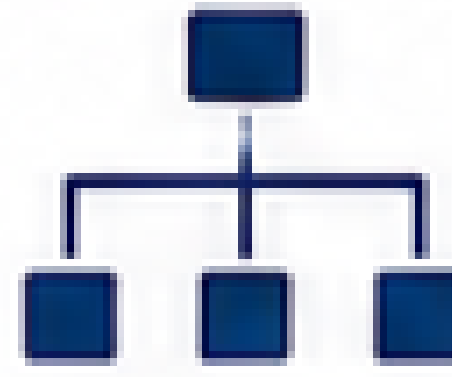
Identify Dominant Sustainability Themes

6 Greenwashing Risk Index Construction



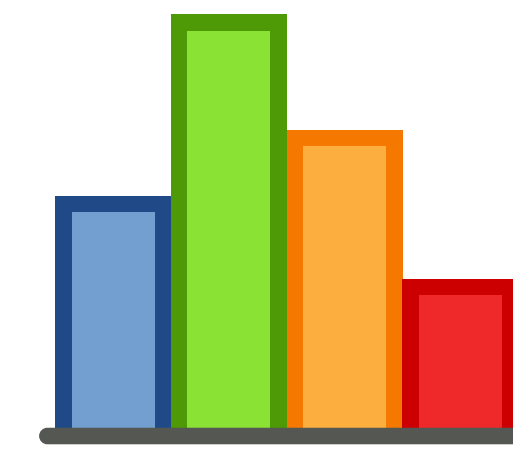
Weighted Composite Index
(0 = Low Risk,
1 = High Risk)

7 ML Modeling



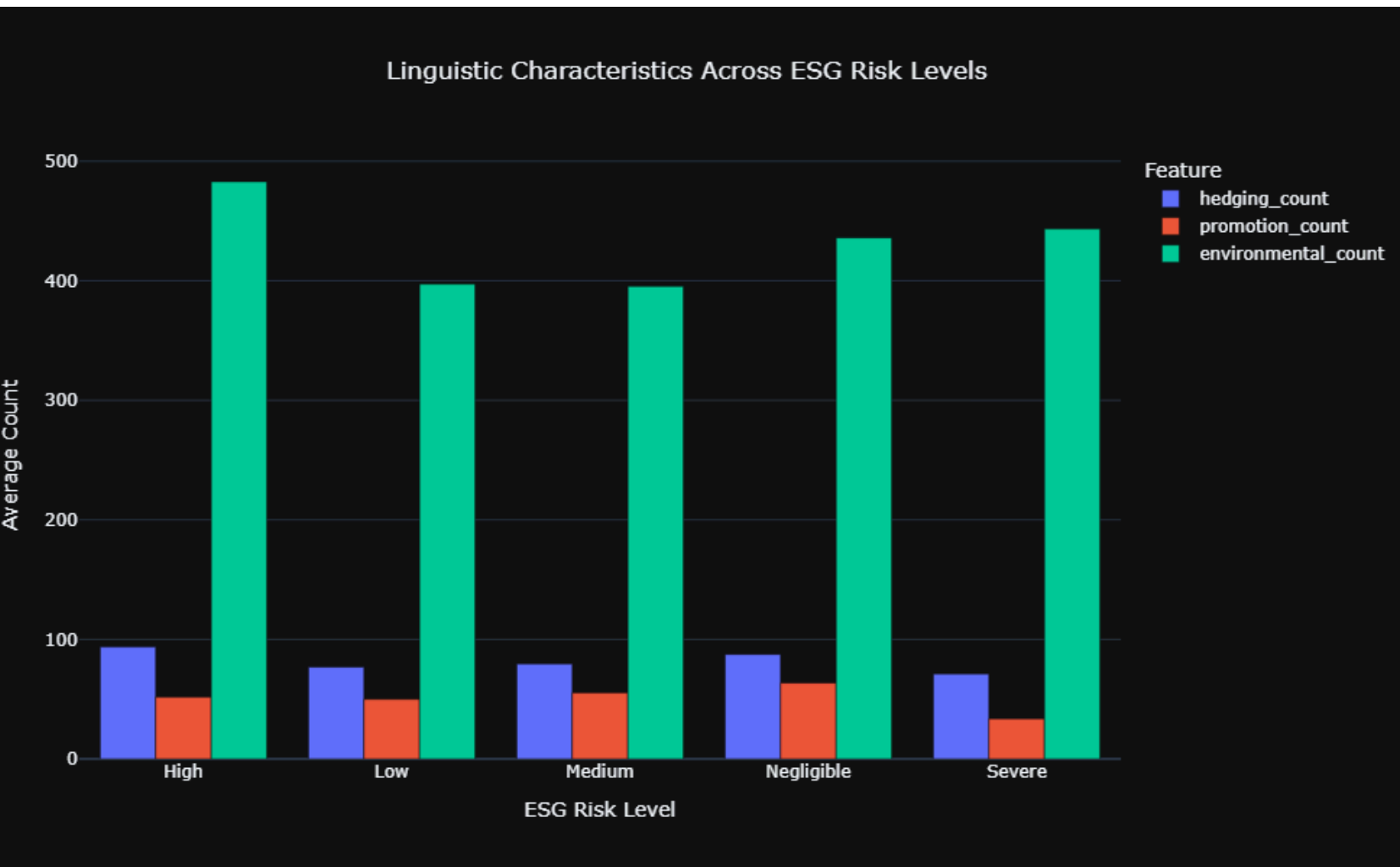
A Random Forest classifier is used for risk prediction.

8 Explainable AI (SHAP)



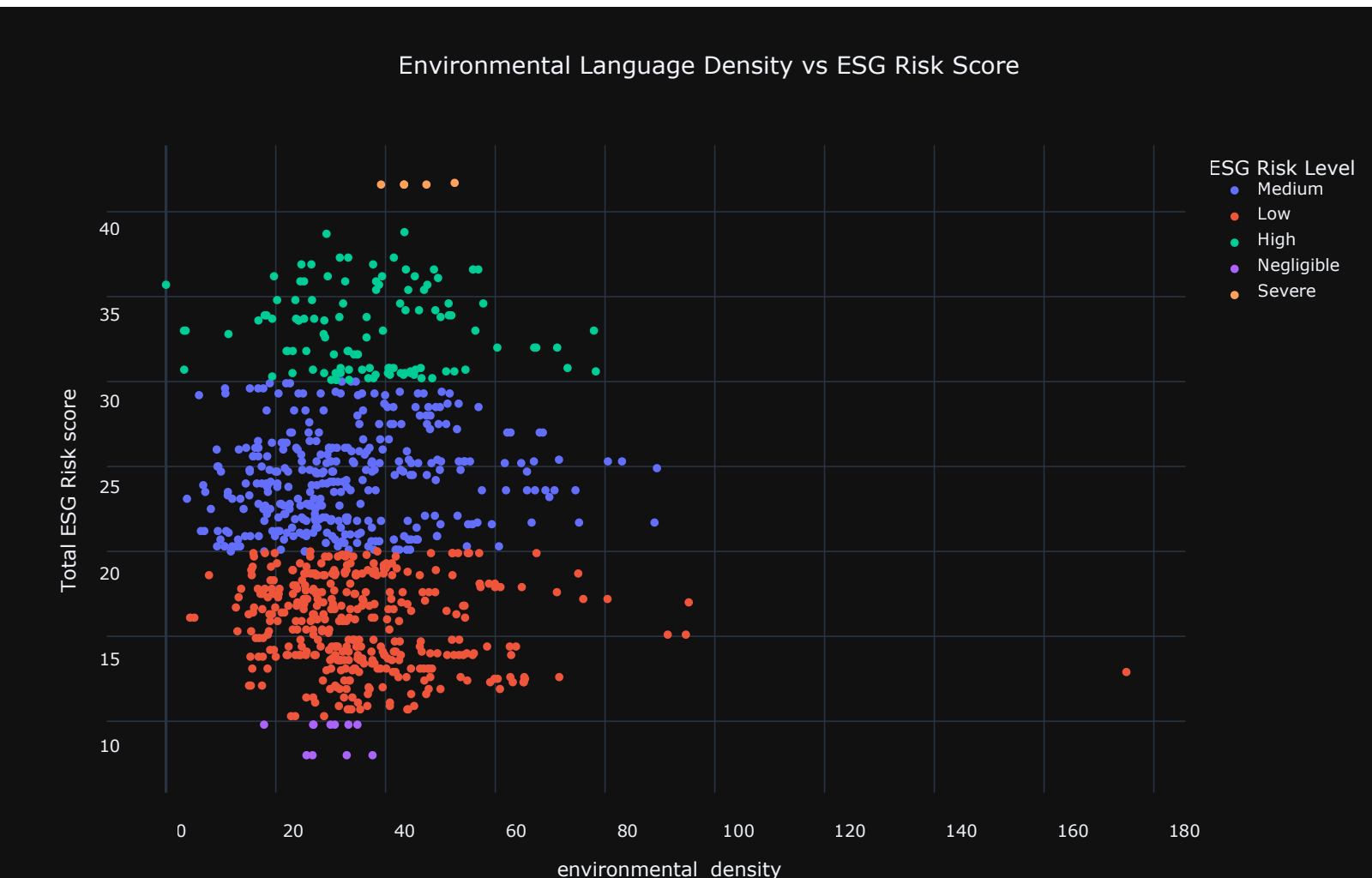
Feature importance analysis is used to identify key drivers of greenwashing risk

RQ1: How do linguistic characteristics vary across ESG risk levels?



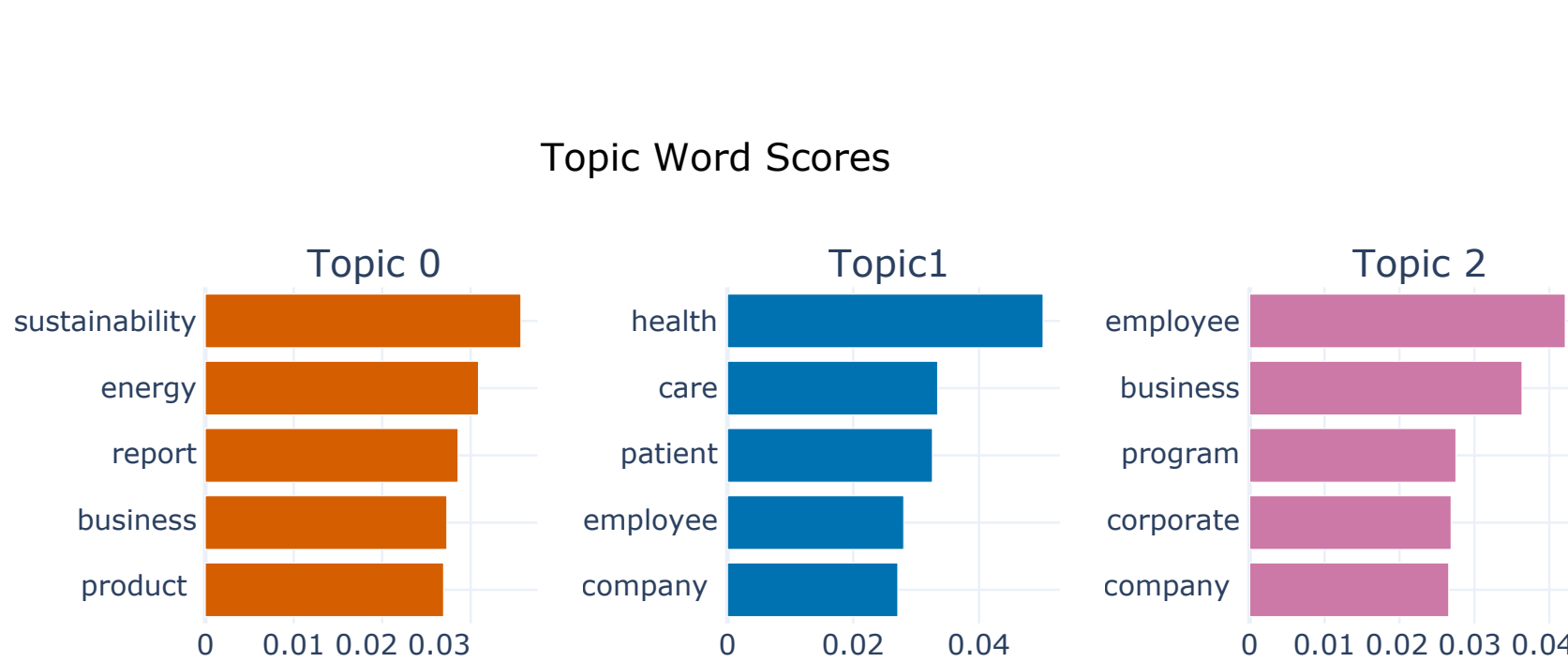
High-risk firms tend to communicate sustainability initiatives more intensively and use greater amounts of cautious or non-committal language.

RQ2: Is sustainability language intensity associated with ESG Risk Scores?



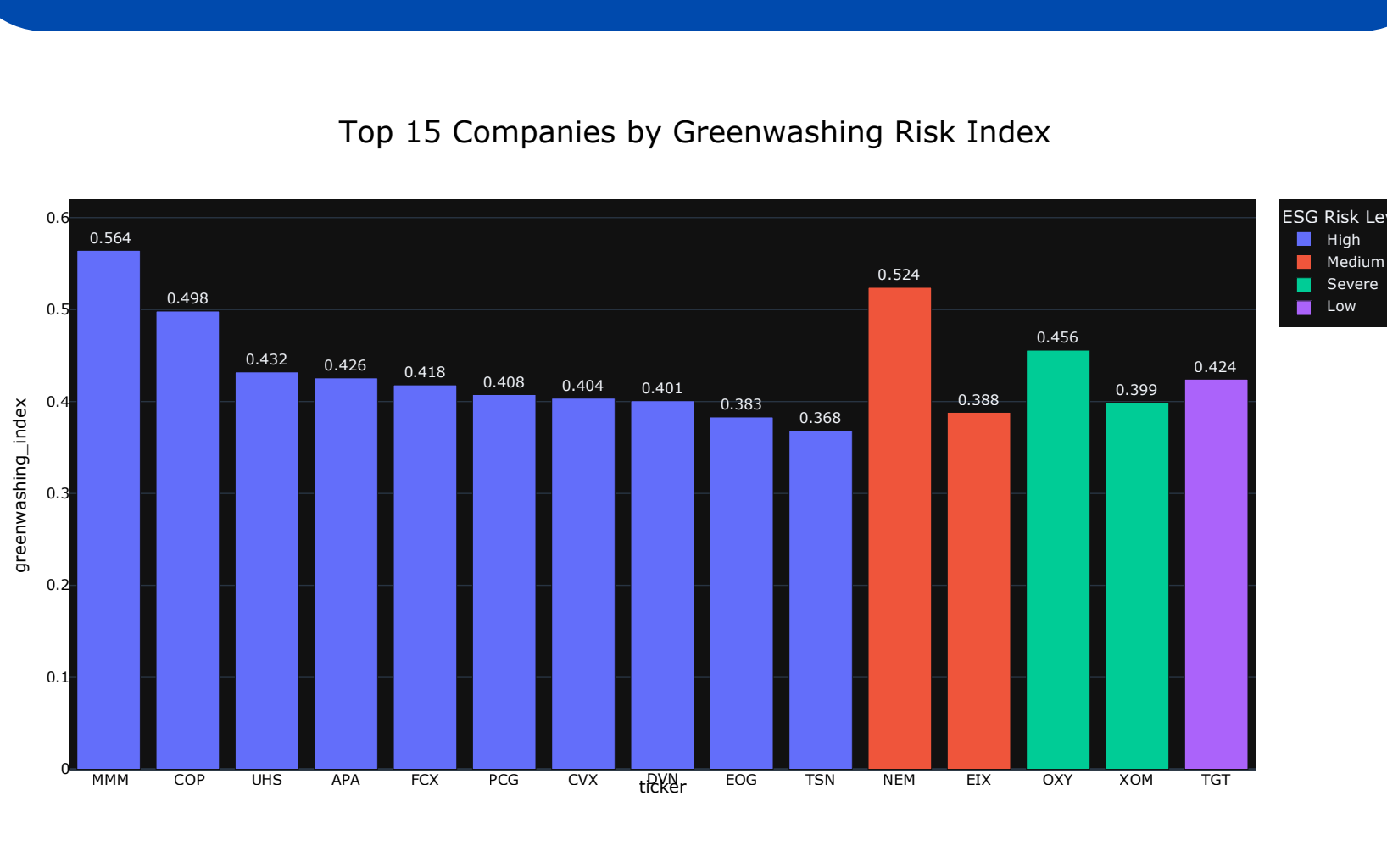
Environmental language density showed virtually no relationship with ESG risk scores. Companies discussed sustainability extensively regardless of their actual ESG performance.

RQ3: What sustainability topics dominate reports of companies with different ESG Risk Levels?



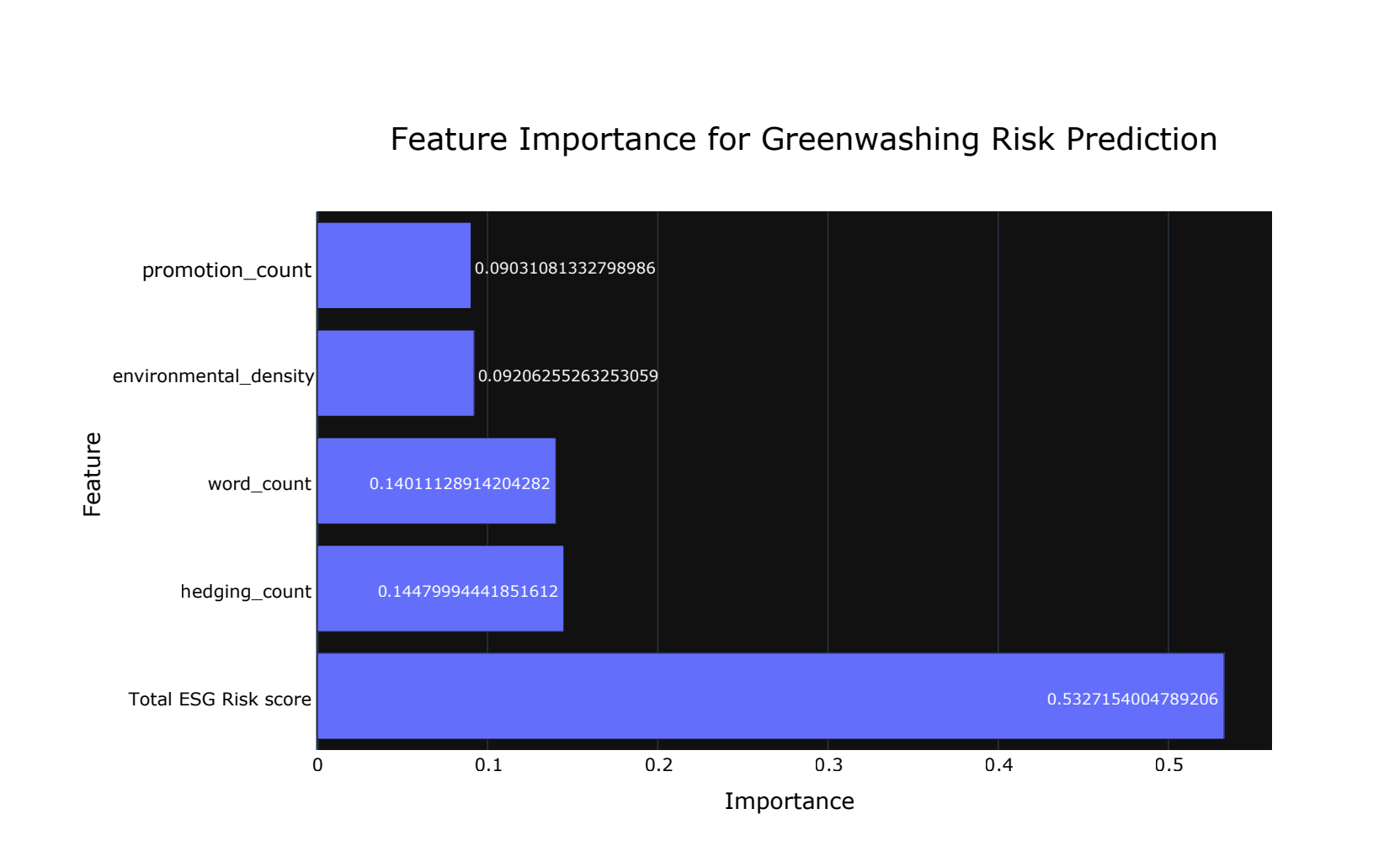
Environmental sustainability and emissions management emerged as the most dominant themes across sustainability disclosures.

RQ4: Can an AI-based Greenwashing Risk Index be developed using linguistic characteristics and ESG risk metrics?



The index successfully identified organisations with elevated potential greenwashing risk by combining sustainability communication patterns with ESG risk indicators.

RQ5: Which linguistic and ESG-related features contribute most to Greenwashing Risk predictions?



Model Performance
(Random Forest Performance)
94.2%

CONCLUSION

This study developed an AI-based framework for detecting potential greenwashing in corporate sustainability reports. By integrating NLP-derived linguistic features, ESG risk metrics, topic modelling, and explainable machine learning, the framework successfully identified disclosure patterns associated with greenwashing risk. The proposed approach provides a scalable and interpretable solution for regulators, investors, and organisations seeking to assess the credibility of sustainability communications.